Causal Inference: Target Trial

- Throughout the course we have thought about trying to use analytical tools to mimic the randomized trial that we would have liked to perform (target trial)
- Framework works well for interventions which you can control (pharmaceutical treatment, surgery, behavioral intervention, income, etc.)
- Framework leaves out many factors which we may be interested in (particularly race/ethnicity, gender, etc.)

A Recent Example

JAMA | Original Investigation

Association of Metabolic Surgery With Major Adverse Cardiovascular Outcomes in Patients With Type 2 Diabetes and Obesity

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Figure 1: JAMA; September 2, 2019

A Recent Example

DESIGN, SETTING, AND PARTICIPANTS Of 287 438 adult patients with diabetes in the Cleveland Clinic Health System in the United States between 1998 and 2017, 2287 patients underwent metabolic surgery. In this retrospective cohort study, these patients were matched 1:5 to nonsurgical patients with diabetes and obesity (body mass index [BMI] ≥30), resulting in 11 435 control patients, with follow-up through December 2018.

EXPOSURES Metabolic gastrointestinal surgical procedures vs usual care for type 2 diabetes and obesity.

MAIN OUTCOMES AND MEASURES The primary outcome was the incidence of extended MACE (composite of 6 outcomes), defined as first occurrence of all-cause mortality, coronary artery events, cerebrovascular events, heart failure, nephropathy, and atrial fibrillation. Secondary end points included 3-component MACE (myocardial infarction, ischemic stroke, and mortality) and the 6 individual components of the primary end point.

RESULTS Among the 13 722 study participants, the distribution of baseline covariates was balanced between the surgical group and the nonsurgical group, including female sex (65.5% vs 64.2%), median age (52.5 vs 54.8 years), BMI (45.1 vs 42.6), and glycated hemoglobin level (7.1% vs 7.1%). The overall median follow-up duration was 3.9 years (interquartile range, 1.9-6.1 years). At the end of the study period, 385 patients in the surgical group and 3243 patients in the nonsurgical group experienced a primary end point (cumulative incidence at 8-years, 30.8% [95% CI, 27.6%-34.0%] in the surgical group and 47.7% [95% CI, 46.1%-49.2%] in the nonsurgical group [*P* < .001]; absolute 8-year risk difference [ARD], 16.9% [95% CI, 13.1%-20.4%]; adjusted hazard ratio [HR], 0.61 [95% CI, 0.55-0.69]). All 7 prespecified secondary outcomes showed statistically significant differences in favor of metabolic surgery, including mortality. All-cause mortality occurred in 112 patients in the metabolic surgery group and 1111 patients in the nonsurgical group (cumulative incidence at 8

Causal Inference: Non-Target Trial

- For such factors, traditional heuristic descriptions of confounding (common causes of treatment and outcome) are not applicable
- Causal methods are useful to create balance among certain characteristics between different groups
- Important to think of what characteristics AT WHAT TIME POINT we want balanced
- Balancing on different characteristics at different time points gives insight on when differences emerge
- For example, ER doctor looking at outcomes by race/ethnicity for those presenting with chest pain can or cannot balance on laboratory values at presentation